

Market Regime Detection and Next-Day Direction Forecasting on Multi-Asset Financial Time Series

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Abstract

This paper applies unsupervised and supervised machine learning techniques to a multi-asset financial time series spanning four major instruments — DXY, Gold, Silver, and S&P 500 — over the trading period March 2024 to May 2026. The dataset, originally constructed for the Data Management course at the University of Milano-Bicocca, integrates price-derived technical indicators with news-based sentiment features. Two problems are addressed: (i) market regime detection via hierarchical Ward clustering and K-Medoids partitioning ($K = 3$), validated through internal indices, external comparison, and a Monte Carlo significance test; and (ii) binary next-day direction forecasting (UP/DOWN) using Decision Tree, Random Forest, Naive Bayes, Logistic Regression, and an Attribute-Selected J48, evaluated under holdout, 10-fold cross-validation, Wilson confidence intervals, and paired t -tests. Results reveal statistically significant but weakly separated cluster structure, while all classifiers fail to exceed the majority-class accuracy baseline of 56.0% — a finding consistent with the Efficient Market Hypothesis.

Keywords: market regime detection, clustering, K-Medoids, Ward linkage, binary classification, CFS, KNIME, financial time series.

1. Introduction

Financial market prediction is among the most studied problems in applied machine learning. The conjecture that historical price data, technical indicators, and sentiment signals carry exploitable information for short-horizon direction forecasting underpins a large body of research in quantitative finance. Yet theoretical arguments — most notably the Efficient Market Hypothesis (EMH) — cast doubt on the systematic predictability of asset returns: in its weak form, the EMH asserts that all information contained in past prices is already reflected in current prices.

This work investigates two complementary research questions on a multi-asset financial dataset:

RQ1: Is there a statistically significant cluster structure in the joint behaviour of DXY, Gold, Silver, and S&P 500, interpretable as distinct market regimes?

RQ2: Do technical indicators and sentiment features carry sufficient information to predict next-day price direction with accuracy meaningfully above the majority-class baseline?

The entire analysis is implemented in KNIME Analytics Platform, structured as a four-section workflow: Preprocessing, Clustering, Classification – Holdout, and Classification – Cross-Validation and Feature Selection.

2. Dataset

The dataset *integrated_data.csv* was personally constructed for the Data Management course at the University of Milano-Bicocca, integrating daily market data for four assets — DXY (US Dollar Index), Gold, Silver, and S&P 500 — with technical indicators computed from OHLCV prices and sentiment scores derived from financial news. The raw dataset contains 2,150 daily observations covering the trading period 27 March 2024 – 15 May 2026. One observation is lost due to the lag operation used to construct the prediction target, yielding a working dataset of **2,149 rows** distributed approximately evenly across the four assets (~537 observations per asset).

Table 1 summarises the 22 columns retained after preprocessing. Following One-to-Many encoding of the categorical variable *asset_name* into four binary dummies (*dxy*, *gold*, *silver*, *sp500*), the modelling feature space comprises **25 predictors**. The binary target *direction* is constructed via the Rule Engine node: *direction* = *UP* if *daily_return_pct(t+1)* > 0, else *DOWN*. The class distribution is moderately imbalanced: **UP: 1,203 (56.0%)**, **DOWN: 946 (44.0%)**, establishing a majority-class baseline accuracy of **56.0%**.

Category	Feature(s)	Type	#
Price / Volume	high, volume, daily_return_pct	Numeric	3
Trend	sma_7, sma_21, ema_7, ema_21	Numeric	4
Momentum	rsi_14	Numeric	1
Bollinger Bands	bollinger_upper, bollinger_lower, bollinger_pct	Numeric	3
Volatility	volatility_7, volatility_21	Numeric	2
Relative Price	price_vs_sma21	Numeric	1
Sentiment	sentiment_mean, sentiment_median, sentiment_std, news_count, positive_ratio, negative_ratio	Numeric	6
News flag	has_news → dummy after One-to-Many	Boolean	1
Asset ID	asset_name → dxy, gold, silver, sp500	String → 4 dummies	4
Target	direction (UP / DOWN)	String	1

Table 1. Feature summary. Columns excluded: ticker, date, open, low, close, daily_return_pct(-1).

3. Preprocessing

The KNIME workflow is structured into four annotated sections, illustrated in Figures 1–4 below.

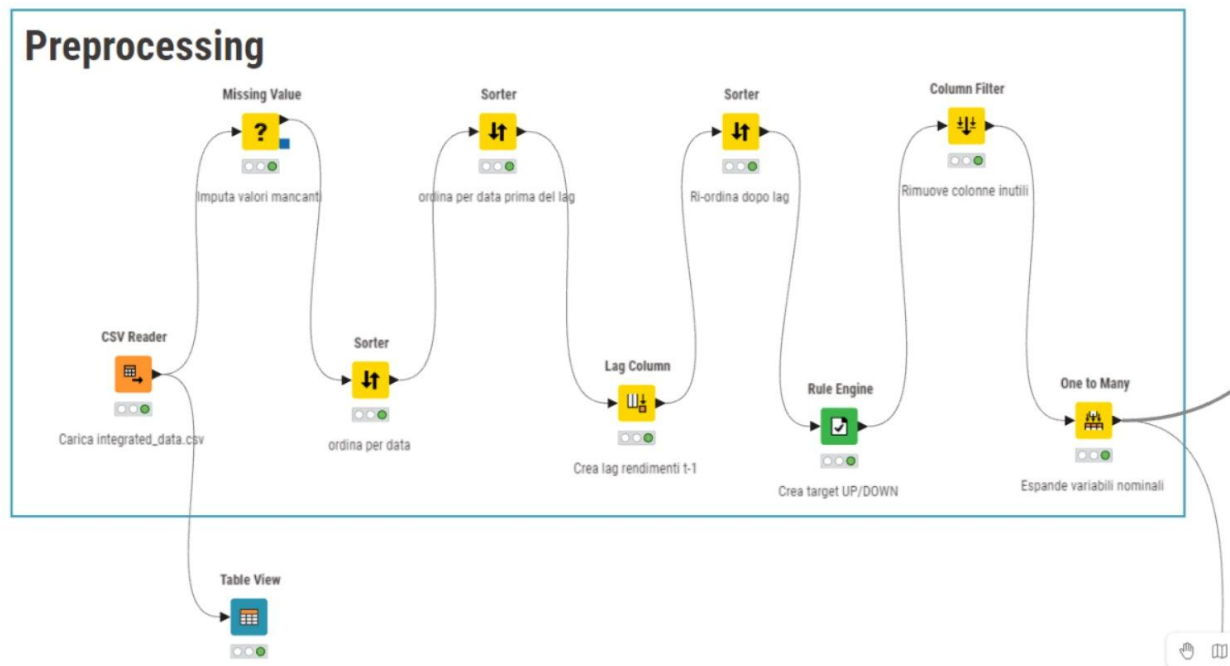


Figure 1. Preprocessing section: CSV Reader → Missing Value → Sorter → Lag Column → Sorter → Rule Engine → Column Filter → One-to-Many. The two output branches feed the Clustering (Z-score) and Classification (Min-Max) normalizers respectively.

As shown in Figure 1, the Preprocessing section ingests the raw CSV via a **CSV Reader** node. A **Missing Value** node handles sporadic gaps with column-mean imputation. Two **Sorter** nodes guarantee chronological ordering by asset and date. A **Lag Column** node shifts *daily_return_pct* forward by one step, creating the next-day return used as the basis for labelling. A **Rule Engine** node converts this value into the binary target *direction*. A **Column Filter** removes raw OHLC prices (open, low, close), the ticker, the date string, and the lagged return column, retaining 22 columns. A **One-to-Many** node encodes *asset_name* into four binary dummies and *has_news* into a numeric indicator. Finally, two **Normalizer** nodes produce separate representations: Z-score standardisation ($\mu = 0, \sigma = 1$) for the clustering branch, and min-max scaling to $[0, 1]$ for the classification branch.

4. Clustering

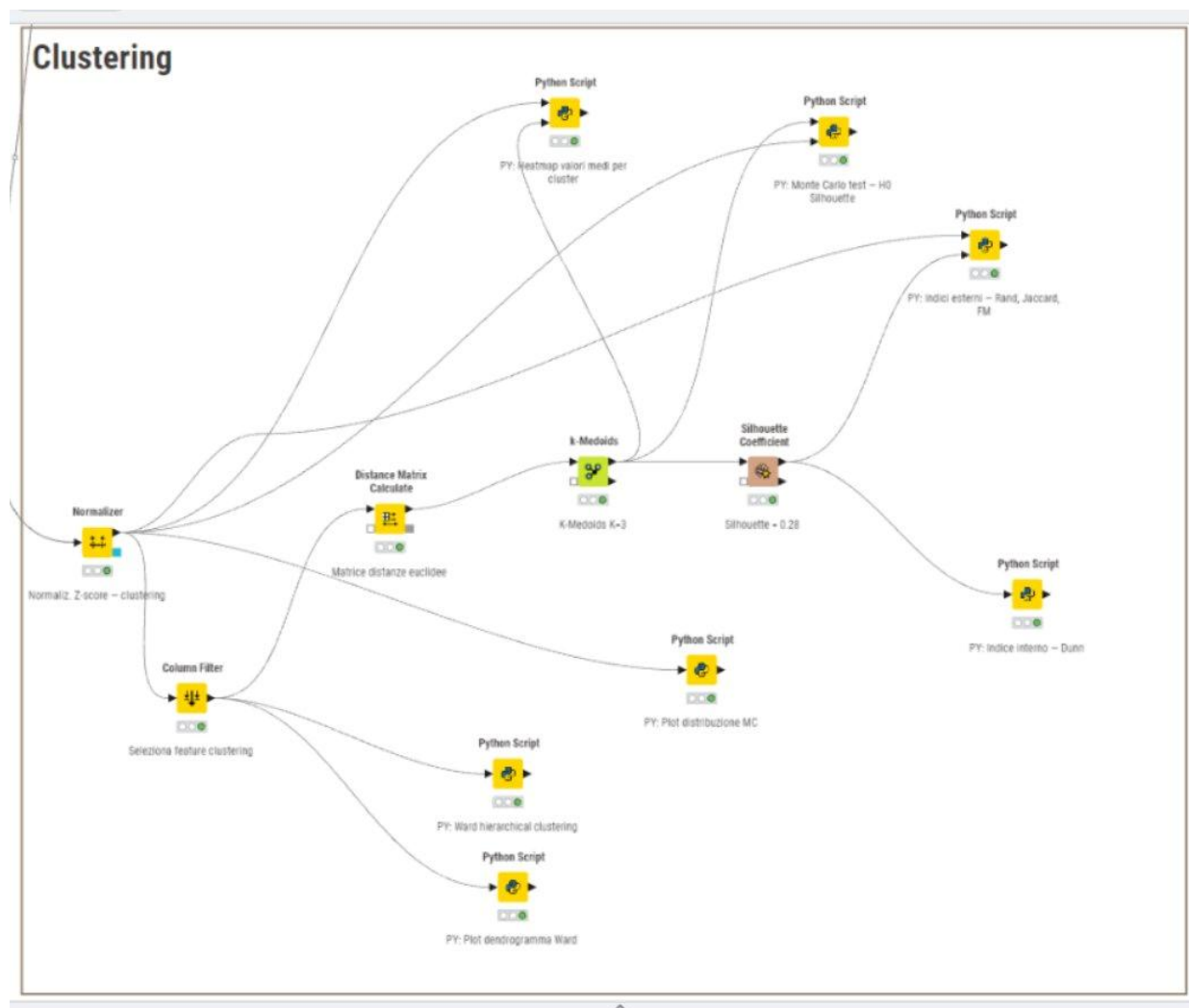


Figure 2. Clustering section: Normalizer (Z-score) → Column Filter (feature selection) → Distance Matrix Calculator → K-Medoids (K=3) → Silhouette Coefficient node; Python Script nodes compute Ward dendrogram, heatmap, Monte Carlo test, Dunn index, and external indices (Rand, Jaccard, FM).

4.1 Cluster Number Selection via Ward Linkage

Ward's minimum variance linkage is applied to the Z-score-normalised data via a Python Script node. The resulting dendrogram reveals a pronounced increase in the merge distance at the three-cluster cut, motivating **K = 3**. The combination of an agglomerative method for K selection with a partitioning method for the final assignment is standard practice.

4.2 K-Medoids Partitioning

A Column Filter node (annotated 'Selezione feature clustering') removes the target variable direction and the One-to-Many encoded asset dummies from the clustering feature space, retaining only continuous numerical indicators for the distance computation. K-Medoids with Euclidean distance and K = 3 is then applied through the native KNIME Distance Matrix Calculator and K-Medoids nodes. K-Medoids is preferred over K-Means because it selects actual data points as cluster centres (medoids), making the solution more robust to outliers — a relevant consideration in financial time series subject to abrupt regime shifts.

4.3 Internal Validation

Two internal indices are computed by a Python Script node. The **Silhouette coefficient** measures cohesion relative to separation; the **Dunn index** measures the ratio of minimum inter-cluster distance to maximum intra-cluster diameter.

$S = 0.28$ — the clusters are discernible but not sharply separated. $D = 0.021$ — the minimum inter-cluster gap is small relative to the largest cluster diameter. These values are characteristic of regime-based financial partitions, where boundaries are inherently gradual rather than discrete.

4.4 External Validation

Ward and K-Medoids assignments are compared via three external indices computed in a Python Script node, treating one partition as the reference and the other as the prediction (Table 2).

Index	Definition (brief)	Value
Rand	Fraction of concordant pairs	0.497
Jaccard	Co-cluster pairs / union of same-class pairs	0.275
Fowlkes-Mallows	Geometric mean of precision and recall on pairs	0.436

Table 2. External validation indices comparing Ward and K-Medoids assignments ($K = 3$).

Values significantly above random agreement (Rand = 0.5 for a random binary partition) confirm structural consistency between the two algorithms, although the moderate Jaccard score indicates some disagreement at cluster boundaries.

4.5 Monte Carlo Significance Test

To assess whether the observed cluster structure is statistically distinguishable from random data, a Monte Carlo test is performed (1,000 permutations, $\alpha = 0.01$). The Silhouette coefficient of the K-Medoids solution ($S = 0.28$) serves as the test statistic. Under H_0 (Random Position Hypothesis), data point positions are equiprobable within the feature space. The empirical p -value is **$p < 0.001$** : no permutation out of 1,000 produces a Silhouette value as large as the observed one. H_0 is rejected, confirming that the three-cluster partition captures genuine structure in the data.

5. Classification

5.1 Holdout Evaluation (70 / 30, Linear Partitioning)

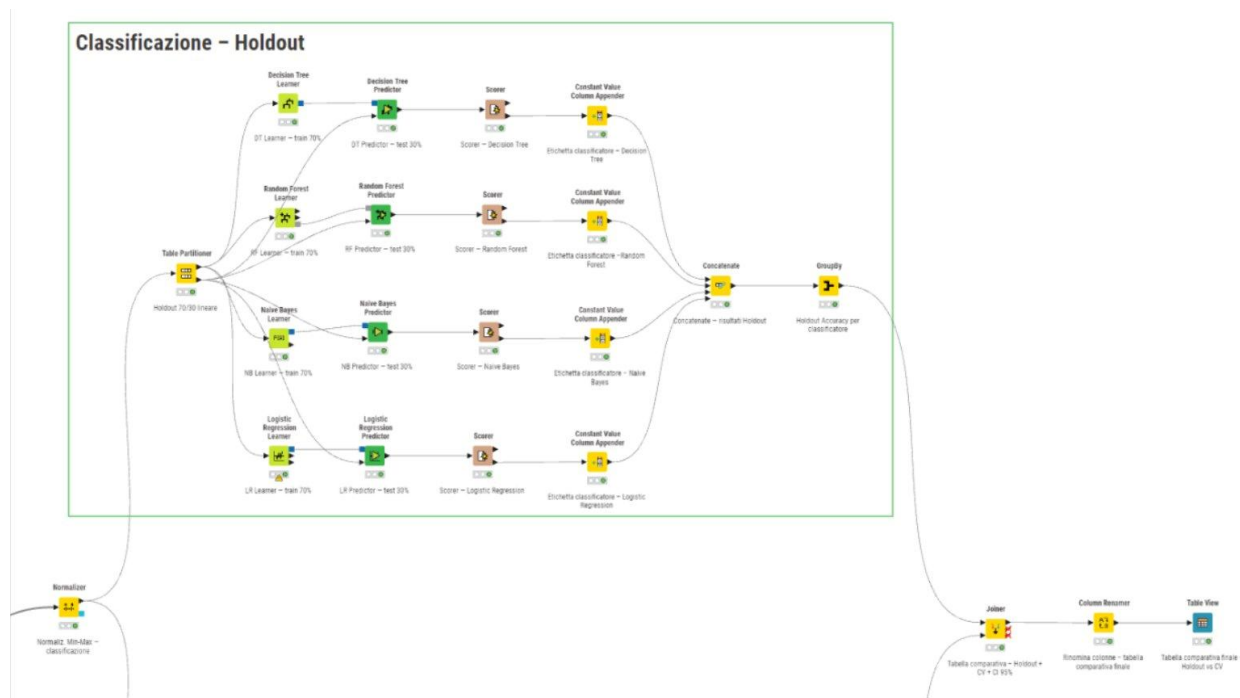


Figure 3. Classification – Holdout section: Table Partitioner (linear 70/30) feeds four parallel Learner → Predictor → Scorer → Constant Value Column Appender pipelines (Decision Tree, Random Forest, Naive Bayes, Logistic Regression); results are collected via Concatenate and aggregated with GroupBy.

A **Table Partitioner** node splits the dataset linearly into 1,504 training observations (70%) and 645 test observations (30%), preserving temporal ordering. Four classifiers are trained and evaluated in parallel: Decision Tree (J48), Random Forest, Naive Bayes, and Logistic Regression. Each pipeline follows the pattern Learner → Predictor → Scorer → Constant Value Column Appender; results are pooled with a Concatenate node and summarised with a GroupBy node. Results are reported in Table 3.

Classifier	Accuracy	Cohen's κ
Decision Tree	0.526	0.029
Random Forest	0.547	0.009
Naive Bayes	0.470	-0.062
Logistic Regression	0.536	-0.003
Majority-class baseline	0.560	—

Table 3. Holdout 70/30 results. All classifiers perform at or below the majority-class baseline (56.0%); $\kappa \approx 0$ throughout.

All four classifiers achieve accuracy in the range [0.470, 0.547], below or at the majority-class baseline of 56.0%. Cohen's κ values lie in [-0.062, 0.029], indicating agreement with a trivial majority-class predictor rather than genuine discriminative learning.

5.2 Cross-Validation and CFS Feature Selection

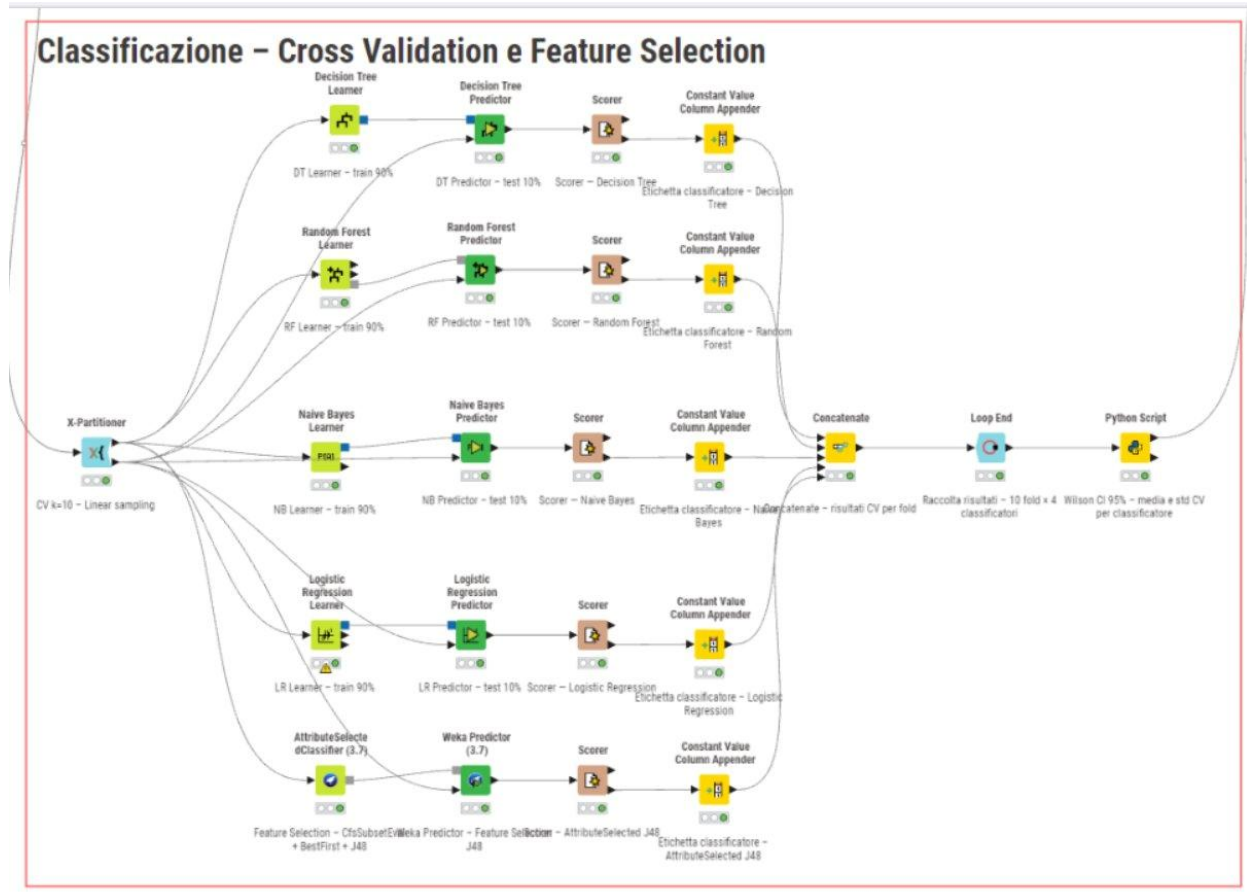


Figure 4. Classification – Cross-Validation and Feature Selection section: X-Partitioner ($k=10$, linear) feeds five parallel classifiers (Decision Tree, Random Forest, Naive Bayes, Logistic Regression, AttributeSelectedClassifier J48); results are pooled with Concatenate, collected by Loop End, and processed by a Python Script node computing Wilson CI and paired t-tests.

A 10-fold cross-validation loop (X-Partitioner / Loop End nodes) evaluates five classifiers: the four above plus an **Attribute-Selected J48**, which wraps J48 (-C 0.25 -M 2) within a CFS evaluator (CfsSubsetEval, BestFirst forward search, stale after 5 expansions). Wilson score confidence intervals at the 95% level and standard deviations are computed by a Python Script node. Results are shown in Table 4.

Classifier	Accuracy $\pm$ SD	κ	CI ₉₅ lower	CI ₉₅ upper
Attr. Selected J48	0.560 $\pm$ 0.033	0.000	0.493	0.625
Decision Tree	0.503 $\pm$ 0.041	-0.007	0.437	0.570
Logistic Regression	0.547 $\pm$ 0.046	0.027	0.480	0.612
Naive Bayes	0.523 $\pm$ 0.048	0.021	0.456	0.588
Random Forest	0.545 $\pm$ 0.049	-0.011	0.478	0.611
Majority-class baseline	0.560	—	—	—

Table 4. 10-fold CV results with Wilson 95% confidence intervals. The majority-class baseline (56.0%) is shown for reference.

The CFS evaluator explores 121 candidate subsets and selects a **single attribute** — *high* — with subset merit = **0.000**. The resulting J48 tree degenerates to a **single leaf** predicting always *UP* (1,935 training instances, 854 misclassified), achieving accuracy 0.560 ± 0.033 , $\kappa = 0.000$ — precisely matching the majority-class baseline. This is the most informative single result of the entire analysis: after a filter-based forward search, **no feature carries any incremental information about next-day direction**.

5.3 Statistical Comparison (Paired t-Test)

Paired t-tests at $\alpha = 0.05$ across the 10 folds identify three statistically significant pairwise differences: *Attr. Selected J48 > Decision Tree*; *Attr. Selected J48 > Naive Bayes*; *Logistic Regression > Decision Tree*. All remaining pairs show no significant difference. Crucially, the apparent superiority of the Attribute-Selected J48 does not reflect genuine predictive power: Decision Tree ($\kappa = -0.007$) and Naive Bayes ($\kappa = 0.021$) perform *below or at* the majority-class baseline, while J48's advantage consists solely in matching it. The confidence intervals of all five classifiers overlap substantially (Table 4), further confirming the absence of meaningful discriminative differences.

6. Limitations and Critical Discussion

Dataset scope. The 26-month window (March 2024 – May 2026) is short by financial modelling standards and encompasses a period of elevated macroeconomic volatility, which may limit the generalisability of both the regime structure and the classification findings.

Sentiment coverage. A substantial fraction of observations carry zero-valued sentiment features (*has_news* = *false*), particularly for DXY. The CFS exclusion of all sentiment-derived features from the selected subset supports the hypothesis that these features contribute noise rather than signal under the current data density.

Target horizon and label construction. The binary label is derived from a one-day-ahead percentage return. At daily resolution, asset returns are empirically close to white noise, making the prediction task inherently difficult. Longer horizons, directional magnitude thresholds, or multi-step targets might yield different conclusions.

Class imbalance. The 56 / 44 split is sufficient to make the majority-class predictor competitive. Cohen's κ provides a more informative performance measure than raw accuracy in this setting; all κ values near zero confirm the absence of genuine learning beyond the prior.

The convergence of all evidence — zero CFS merit, single-leaf J48, $\kappa \approx 0$ across five classifiers and two evaluation protocols, Wilson confidence intervals entirely overlapping the 56.0% baseline — constitutes strong empirical support for the **Weak Form of the Efficient Market Hypothesis** for the assets and time window examined.

7. Conclusions

This paper has presented a KNIME-based machine learning study on a multi-asset financial time series. The **clustering analysis** identifies three market regimes that are statistically non-random (Monte Carlo $p = 0$), though their internal separation is modest (Silhouette = 0.28, Dunn = 0.021), reflecting the continuous and overlapping nature of financial market states. The moderate external indices (Rand = 0.497, Jaccard = 0.275, Fowlkes-Mallows = 0.436) confirm partial structural agreement between the Ward and K-Medoids algorithms.

The **classification analysis** provides unambiguous evidence that the 25 technical and sentiment features constructed from publicly available price data carry no exploitable predictive signal for next-day direction under the experimental conditions examined. All five classifiers achieve accuracy at or below the 56.0% majority-class baseline, with $\kappa \approx 0$ throughout. CFS reduces the feature space to a single attribute (*high*) with zero merit, and the resulting J48 tree is a trivial single-leaf predictor — the strongest possible null result for a classification system.

These findings are consistent with the Efficient Market Hypothesis and underscore the intrinsic difficulty of short-horizon return prediction from publicly available indicators. Future work could explore non-linear feature interactions via deep learning architectures, tick-level order-book features, alternative labelling strategies, or ensemble methods operating over multiple time horizons.